INCH-POUND

MIL-C-24643/41B
22 November 1994
SUPERSEDING
MIL-C-24643/41A
14 March 1994

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, TYPE LS1SAU
(INCLUDING VARIATION LS1SA)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-24643.

This cable can be manufactured in two variations: LS1SAU (unarmored), and LS1SA (armored).

REQUIREMENTS:

Qualification required.

Construction (watertight)

First - Copper conductor, uncoated, ASTM B 286, size 20-7. maximum diameter 0.039 inch.
Second - Separator may be used at manufacturer's option where required to provide free-stripping insulation.
Third - Crosslinked polyethylene insulation. Minimum wall thickness 0.015 inch; wall thickness may vary from nominal as necessary, in order for completed cable to meet the specified electrical requirements.
* Fourth - Braided shield of AWG No. 34 or No. 36 uncoated copper. Braid angle of 30 to 35 degrees minimum coverage of 85 percent.
Fifth - Shield insulation of two polyester tapes, type G in accordance with MIL-I-631, outer tape sealed, an alternate shield insulation may consist of one tape, type G of MIL-I-631, plus an extruded jacket of a suitable transparent material, minimum average thickness of the transparent material 0.003 inch. The standard identification code shall be applied by method 2 on the inner tape.
Sixth - The required number of singles (see table I) cabled together with a lay of not greater than 24 times the pitch diameter of the layer. Cabling sequence shall be consecutive, starting with no. 1, from the center outward.
Seventh - Binder tape, applied helically with overlap.

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Construction (watertight) (continued)

Construction variant for type LS1SAU

Eighth - Cross-linked polyolefin jacket. Minimum average thickness 0.063 inch.

Construction variant for type LS1SA

Eighth - Cross-linked polyolefin jacket. Minimum average thickness 0.063 inch.
Ninth - Braided metal armor

TABLE I. Details.

Military 1/ part no. M24643/41	Size 2/	Number of Conductors	Overall diameter (inches)			
			TYPE LS1SAU		TYPE LS1SA	
			min	max	min	max
-01UN	44	44	0.910	0.990	0.960	1.040

1/ Unarmored. When armor option is required, see MIL-C-24643 for configuration of part number.
2/ Type LS1SAU and LS1SA.

EXAMINATION AND TESTS:

	Requirements
Basic electrical:	
Conductor Resistance - ohms/1000 feet at 25°C, maximum.....	10.71
Voltage withstand - volts, root mean square, minimum	
Conductor to shield	2000
Shield to shield	500
Insulation resistance - Megohms/1000 feet, minimum	
Conductor to shield	500
Component shield to shield	100
Conductor and shield continuity	No failure
Group A:	
Visual and dimensional	No failure
Capacitance - 1 kilohertz (KHz), pF/ft, maximum	72
Group B:	
Cross-linked proof test (percent, maximum)	
Insulation	50
Jacket (When tested at 200°C)	50
Tear - pounds per inch thickness, minimum (ASTM D 470)	35
Physicals (unaged)	
Insulation	
Tensile strength - lb/in^2, minimum.....	1800
Elongation - percent, minimum	250

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EXAMINATION AND TESTS (continued):

	Requirements
Jacket (cable)	
Tensile strength - lb/in^2, minimum	1300
Elongation - percent, minimum	160
Group C:	
Physicals (aged)	
Insulation	
Air oven	
Tensile strength - percent of unaged, minimum	80
Elongation - percent of unaged minimum	80
Jacket (cable)	
Air oven	
Tensile strength - percent of unaged, minimum	60
Elongation - percent of unaged minimum	60
Hot oil immersion	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Shrinkage	No failure
Heat distortion - percent of unaged, maximum	30
* Permanence of printing (jacket) - cycles, minimum	125
Cable sealant removability	No failure
Armor (type LS1SA only) - conformance to material, construction and coverage	No failure
Shield - conformance to material, construction and coverage	No failure
Group D:	
Flame propagation (cable)	No failure

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

	Requirements
Aging and compatibility (cable) (125 ± 5°C)	No failure
Abrasion resistance (jacket) - scrapes, minimum	75
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
Shield insulation	2
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2

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QUALIFICATION INSPECTION (continued):

	Requirements
Immersion (jacket)	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Smoke index, maximum	
Jacket	25
Fillers	45
Insulation	45
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80
Weathering (jacket)	No failure

UNIT ORDERING LENGTHS: 1000 feet nominal.

NOTE: The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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